

Fabio Taccani

M.Sc. Candidate in Theoretical Physics, University of Padua

Master's thesis researcher at Heidelberg University

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Research Interests

Conformal field theory and conformal bootstrap; analytic and numerical consistency methods in quantum field theory; Virasoro and current algebras; modular invariance; operator-theoretic aspects of quantum field theory; vertex operator algebras; topological defects and generalised symmetries.

Education

2024–2026 **University of Padua** – Padua, Italy

M.Sc. in Physics, Fundamental Interactions; expected completion: September 2026

Master's thesis: *A Modified Fock-Space Quantisation of the Bosonic String*

Supervisors: Prof. Johannes Walcher, Heidelberg University, and Prof. Roberto Volpato, University of Padua

2025–2026 **Heidelberg University** – Heidelberg, Germany

Erasmus exchange and Master's thesis research

2021–2024 **University of Padua** – Padua, Italy

B.Sc. in Physics

Bachelor's thesis: *Classical Electrodynamics of Massive Gauge Fields*; supervisor: Prof. Kurt Lechner

Research Experience

2025–2026 **Master's Thesis Research: Modified Bosonic-String Quantisation**

Heidelberg University and University of Padua

Supervisors: Prof. Johannes Walcher and Prof. Roberto Volpato

- Constructed a positive-definite kinematical Fock representation of the bosonic string by modifying the polarisation of the timelike oscillator sector.
- Derived the associated Virasoro representation, including its modified central charge and normal-ordering constant, and analysed the resulting departure from the standard covariant quantisation.
- Proved that target-space Lorentz boosts are not unitarily implementable in this representation, using both generator-domain and Bogoliubov-transformation arguments.
- Proved, for the maximal closed realisations considered in the manuscript, that the joint kernel of all positive Virasoro modes is trivial in the kinematical Hilbert-space completion for every momentum and every finite dimension $D \geq 2$. Consequently, the corresponding physical Hilbert space is trivial, independently of the zero-mode constraint.

Research Experience

2024–2025 **Research Internship: Topological Defects in the Monster Module**

University of Padua; supervisor: Prof. Roberto Volpato

- Studied topological defects and twisted sectors in the Monster conformal field theory for $k = 3, 5, 7$.
- Computed modular S -matrices and solved modular-consistency equations for defect and twisted-sector partition functions.
- Derived the corresponding fusion rules and analysed their compatibility with the underlying conformal-field-theory structure.

Manuscript in Preparation

- 2026 **F. Taccani**, *Lorentz Non-Implementability and Triviality of the Physical Hilbert Space in a Positive Bosonic-String Fock Representation*.

Computational Skills

Mathematica: symbolic calculations, algebraic manipulation, and analytical checks.

Talks and Presentations

- 2025 **Topological Defects in the Monster Module**. Contributed short talk, Gong Show, *The Physics and Mathematics of Boundaries, Impurities, and Defects*, Isaac Newton Institute, Cambridge.

Selected Schools, Workshops and Conferences

- 2026 **CIRM**, Marseille – accepted participant in the Research School *Loop Models, Conformal Field Theory, and Geometry* and the conference *Geometric and Probabilistic Aspects in Quantum Field Theory*.
- 2026 **All Around Integrability II**, Bologna – workshop participant.
- 2026 **CERN Winter School on Supergravity, Strings and Gauge Theory**, CERN, Geneva.
- 2025 **The Physics and Mathematics of Boundaries, Impurities, and Defects**, Isaac Newton Institute, Cambridge.
- 2025 **Petnica Summer School on String Theory**, Petnica Science Center, Serbia.

Teaching Experience

- 2024–2025 **University Tutor, University of Padua**: Computational Laboratory, B.Sc. in Materials Science; General Physics, B.Sc. in Biotechnology.

Awards and Academic Service

- 2024–2026 **Fondazione Franchetti Merit Scholarship**, awarded in 2024, 2025, and 2026.
- 2024–2026 Student Representative, Department of Physics and Astronomy, University of Padua.